A Resource Allocation Strategy in Internet of Vehicles Based on Multi-Task Federated Learning and Incentive Mechanism

Jianquan Zhang, Fangting Huang, Shuqing Zhu, and Xiao Xiao

***Abstract*— With the continuous emergence of Internet of Vehicles (IoV) applications, the demand for computational resources of many resource-intensive applications in IoV has shown an explosive growth trend, which poses a serious challenge to the limited computational resources of the vehicles themselves. This paper designs a federated learning structure with a two-layer game for vehicular networks, using intelligent roadside terminals for federated optimization. Meanwhile, this paper proposes a Federated Learning and Cloud-Edge Gaming with Incentive-Driven (FL-CEGID) algorithm for dynamic task offloading in IoV. Our proposed algorithm optimizes vehicle and computing resource allocation as well as cache updates through a hierarchical distributed approach, which has separate vehicle and edge intelligence strategies for offloading decisions and caching strategies. The experimental results show that our proposed FL-CEGID has significant improvements in transmission capacity, transmission delay, and advantages in different key tasks and times in IoV compared to other schemes.**

***Index Terms*— Internet of Vehicles, federated learning, incentive mechanisms, cloud-edge game, resource allocation.**

I. Introduction

With the development of automation, mobile communication technology and artificial intelligence, vehicles are evolving in the direction of automation, networking and intelligence, forming intelligent connected vehicles. Intelligent connected vehicles organically combines Internet of Vehicles (IoV) with intelligent vehicles, carrying sensing, video processing, communication, intelligent control and pattern recognition equipment to perceive the vehicle's own state and the surrounding environment, and obtain comprehensive information about the entire transportation and communication network. Thus, the direction and speed of the vehicle can be controlled, and the autonomous driving of the vehicle can be realized with the help of "human-vehicle-road-cloud" cooperative system.

Federated learning is a distributed machine learning method in "client-server" mode, which can realize collaborative learning on a large amount of data without sharing the original data among data owners. The client locally trains the learning model and only uploads the model parameters to the server for aggregation. In this way, the computational power of multiple learning agents can be fully utilized to improve the learning efficiency, while safeguarding the privacy and security of user data. Compared with cloud computing and Mobile Edge Computing (MEC), federated learning does not need to share the raw data on the client with the server, so it provides a guarantee for the client's data privacy. Compared with the transmission of raw data, federated learning effectively reduces the communication overhead of IoV [1], [2]. IoV is an important scenario for federated learning. Emerging IoV systems involve a large amount of vehicle sensor data and a variety of smart driving and intelligent transportation applications in complex vehicular environments [3]. At the same time, some emerging services (e.g., connected smart driving and intelligent transportation systems) require unprecedented high reliability, high accuracy, and fast response. Some services have extremely different resource requirements in terms of time, location, and content on different users. IoV systems that consider only the intelligent processing capabilities of individual vehicles or rely on transferring vehicle data to the cloud to be processed by centralized cloud servers will not be able to meet the demands of emerging services [4], [5].

Federated learning can enhance autonomous driving and intelligent traffic management by enabling collaborative model training without sharing raw data. In autonomous driving, vehicles process sensor data like camera images and radar signals to navigate and make decisions. Through federated learning, vehicles can collaboratively improve object detection and path planning models, with edge servers coordinating the aggregation of local model parameters into a global model for real-time vehicle decision-making. This improves autonomous driving safety and data privacy. In order to cope with the

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above challenges, this paper analyzes the federation learning training process, optimizes and improves its client selection, server-side global model aggregation and client-side local model training, and designs an adaptive federation learning framework and algorithm for edge intelligence. Through the above work, this paper realizes the optimization and improvement of the federated learning algorithm for edge intelligence, and provides a new solution for the implementation of edge intelligence deployment.

The main contributions of this study are outlined as follows:

(1) This paper designs a federated learning structure with a two-layer game for vehicular networks, using intelligent roadside terminals for federated optimization.

(2) This paper proposes a Federated Learning and Cloud-Edge Gaming with Incentive-Driven (FL-CEGID) algorithm that combines federated learning with cloud edge gaming to optimize vehicle offloading, resource allocation, and caching strategies using state, action, reward, and edge agents.

(3) This paper constructs a multi-task federated learning and analysis incentive mechanism, which incentives data owners through reward rates, optimizes the allocation of computing resources, and makes task offloading decisions.

The rest of this paper is organized as follows. Section II discusses related work. Section III presents the federated learning hierarchical architecture between multiple intelligent roadside terminals for IoV. Section V discusses the joint incentive-driven learning algorithm based on cloud-edge game. Section V shows the simulation experimental results. Section VI concludes the paper with a summary and future research works.

II. Related Work

This section will review the research work related to federated learning and the IoV, focusing on the following key areas: the application and challenges of federated learning, the construction of collaborative learning and the infrastructure of the IoV, the application of deep reinforcement learning in IoV.

Federated learning enables vehicles to learn from distributed data while safeguarding user privacy, but faces challenges like data and node resource heterogeneity. Federated learning's reliance on multiple aggregation rounds also strains communication and computational resources, increasing costs for vehicle users. The disparity in vehicle resources can delay model updates, affecting service efficiency. Therefore, improving local data quality and strategic participant selection are crucial to enhance federated learning in IoV [6]. In order to realize a more intelligent IoV system, there is an urgent need for collaborative learning between different vehicles and RSUs to achieve the purpose of utilizing the data, resources and capabilities of more intelligent vehicles, and at the same time, to satisfy the requirements of privacy protection of the vehicle users deploying federation in IoV learning, in general, mobile vehicles are used as clients of federated learning, and federated learning aggregation servers are deployed on RSUs or base stations [7]. In some studies, a two-tier federated learning structure is considered: in addition to the low-tier learning between RSUs and a number of vehicles within their range, a high-level federated learning is constituted between the cloud and a number of RSUs, with the aim of aggregating a larger range of data information and obtaining a more effective machine learning model. In the realms of connected vehicles and federated learning, research has tackled issues like resource allocation, task offloading, and data privacy. Initially, the work focused on establishing federated learning infrastructure and algorithms [8]. The strategy included improving the federated learning algorithm by enhancing client selection and model aggregation to achieve better model accuracy and faster convergence speed [9]. Later, researchers explored the combination of edge computing and deep learning to optimize task unloading and resource allocation [10], [11].

Several research works have been done to improve the performance of vehicular networking using mobile edge computing and deep reinforcement learning. Zhang et al. [12] used deep reinforcement learning to develop an intelligent flow control scheme to solve the joint optimization problem and allocate network resources efficiently. He et al. [13] proposed an improved deep reinforcement learning algorithm to improve the satisfaction of QoE and to find the optimal offloading pattern saving energy consumption. Qi et al. [14] proposed a service offloading decision framework in vehicle networking to solve the resource scheduling problem of offloading decision, considering long term planning, and using deep reinforcement learning to obtain the optimal solution for multiple task offloading decisions. To solve the problem of real-time vehicle response, Lee et al. [15] proposed a heuristic algorithm to find an effective solution and combined it with reinforcement learning to make more effective resource allocation decisions using collected vehicle movement and parking states. In order to cope with the complex and changing vehicular environment, Zhang et al. [16] integrated digital twin and artificial intelligence into the vehicular edge computing network, scheduling computational task offloading and edge resource allocation through multi-intelligent deep reinforcement learning. Some scholars have done research on communication stability and resource allocation in vehicular networking. Lin et al. [17] proposed a content-aware stable multimedia communication algorithm for vehicular networking, which adjusts the data relevance of vehicles within a certain range to maximize data sharing and achieve a stable network structure for multimedia communication. In order to solve the problem of resource allocation, Sun et al. [18] proposed an optimal resource allocation method based on federated learning for optimal resource allocation, assigning the best CPU cycles to high priority tasks, optimizing the delay energy product, balancing the delay and energy performance, then estimating the execution time of edge servers by using federated learning. Nguyen et al. [19] modeled data redundancy and collaborative task computation, using submodule optimization techniques to design effective algorithms to minimize the amount of data transfer, followed by proposing collaborative task computation scheme to optimize the cost and guarantee the latency. Naeem et al. [20] presented a scheduling algorithm that leverages software-defined networks for intelligent scheduling of action value distributions. This algorithm employs generative adversarial networks and introduces a reward pruning technique to guarantee stable training of these networks.

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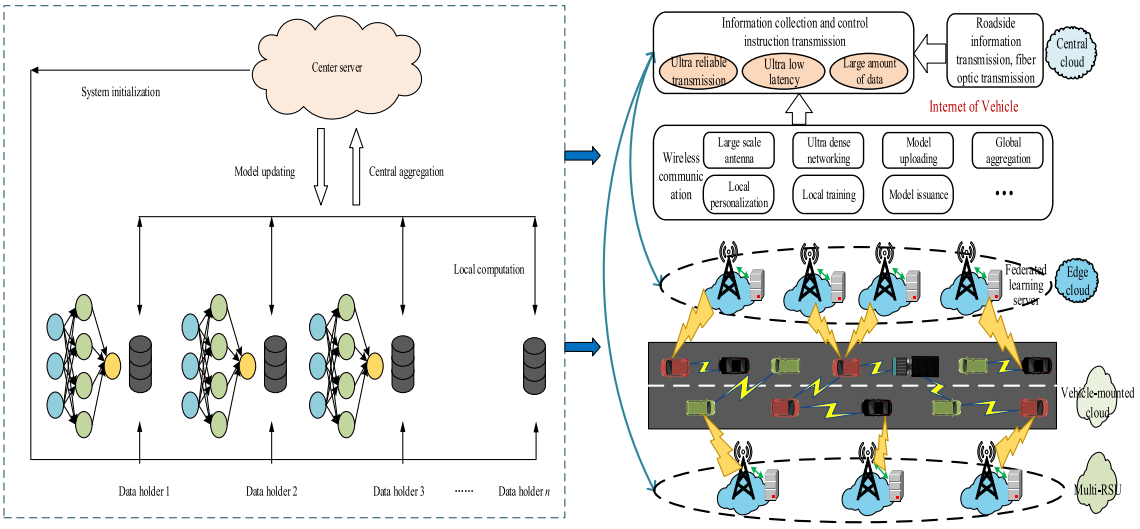


Fig. 1. Federated learning hierarchy among multiple intelligent roadside terminals in IoV.

Regarding the application of federated learning to vehicular networks, Samarakoon et al. [21] proposed a distributed solution based on federated learning for the problem of transmitting power control and resource allocation of vehicles in urban road environments, where the vehicles locally train their own model parameters at small scales, and then globally through model averaging via roadside units at large time scales. Ye et al. [22] considered the image training problem based on edge computing in in-vehicle applications, where each vehicle locally trains Deep Neural Network (DNN), and in order to solve the data silo problem of each vehicle, model sharing is accomplished based on federation learning. Zhang et al. [23] combined blockchain and federation learning for the consideration of the security and privacy of the edge computing of the IoV, and through the directed acyclic graph to represent the licensed blockchain. Chen et al. [24] solved the problem of drone-assisted edge cache replacement policy, and at the user side, based on the echo state network to predict the user's movement pattern and request distribution based on the user's movement context information. He et al. [25] solved the problem of deploying federated learning algorithms in edge intelligence for security and privacy considerations. He et al. [26] solved the problem based on semi distributed reinforcement learning, which is based on semi distributed reinforcement with high communication costs and data leakage risks. For the problem of minimizing task offloading latency, Chen et al. [27] regularly involved all users and edge sides in global aggregation, and abstracted the latency minimization problem as a bilateral many to one matching problem. Regarding the time-varying characteristics of user preferences, Zheng et al. [28] adopted a Long Short-Term Memory (LSTM) network based replacement strategy to predict user preferences on the user side, and obtained the aggregation degree on the edge side by weighting the global aggregation with the loss function of the current model.

This paper enhances federated learning by allocating each RSU server a unique cloud model, improving personalization through attention to detail and local data characteristics. The proposed FL-CEGID designs with a hierarchical structure and cloud-edge game theory, optimizes vehicle and computing resource allocation, addressing issues of practicality, scalability, and privacy in vehicular networks.

III. Federated Learning Hierarchical Architecture Between Multiple Intelligent Roadside Terminals for IoV

In the IoV federated learning hierarchy, synchronized learning between the cloud server and RSU servers is influenced by vehicle mobility, causing learning disparities across RSU coverage areas. This results in varying learning levels among RSU servers, which leads to models with improved personalization and performance after cloud server aggregation.

A. Layered Structure

In the hierarchical structure of federated learning among multiple RSUs, the lower level is the federated learning structure between several vehicle clients and their RSU servers in each single RSU, and the higher level is the federated learning structure between several RSU servers and cloud servers. The high-level learning adopts the synchronous federated learning mode, i.e., every time after a fixed time interval, all RSU servers and cloud servers carry out the federated learning rounds of the "upload - aggregate - download -personalization" process. Due to the mobility of vehicles in low-level learning, the federal learning situation within different single RSUs is not the same, resulting in different levels of model learning on different RSU servers at the same time; therefore, in high-level learning, the learning level of models uploaded to the cloud server by different RSU servers will also be different. The overall structure is shown in Fig. 1.

Fig. 1 illustrates the federated learning process, beginning with local computation and extending to the integration of

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Edge cloud and IoV to enhance intelligent vehicle management and traffic safety. The process starts with multiple data holders (data holder 1 to data holder n), each responsible for storing, managing, and locally processing their data. After local computation, these data holders send their updated models to a central server, which aggregates these updates, processes them, and initializes the system with the aggregated knowledge, thereby providing feedback to the data holders. Simultaneously, the relationship between the edge cloud and IoV unfolds through sequential steps starting with the collection and control of information via large-scale communication facilitated by wireless antennas from vehicles and roadside devices. This structured approach ensures local data processing, central or edge cloud aggregation, and model updating through federated learning.

In the federated learning high-level architecture between multiple RSUs and the cloud, one federated learning server is deployed on each RSU, and the set of servers on a number of RSUs is denoted as$R = \{r_1, r_2, \ldots, r_n\}, n > 0$. In round$p$, the model on the RSU server r_n is denoted by $w_{r_n}^p$. The attention information carried by the model is denoted as $\xi_{r_n}^p$, the model accuracy as $acc_{r_n}^p$, and the model loss as $loss_{r_n}^p$. The attention information is used to focus on the key information of the task, and the precision and loss are used to represent the learning level of the model.

In round p, each RSU server r_n uploads its model $w_{r_n}^p$ to the cloud server r_{cloud}, so the cloud server r_{cloud} will receive the model collection$\left\{w_{r_1}^p, w_{r_2}^p, \ldots, w_{r_n}^p\right\}, n > 0$. The corresponding attention information collection is$\left\{\xi_{r_1}^p, \xi_{r_2}^p, \ldots, \xi_{r_n}^p\right\}$, the model accuracy collection is$\left\{acc_{r_1}^p, acc_{r_2}^p, \ldots, acc_{r_n}^p\right\}$, and the model loss collection is$\left\{loss_{r_1}^p, loss_{r_2}^p, \ldots, loss_{r_n}^p\right\}$.

Unlike the traditional federated learning algorithms where only a single global model is computed at the global server, a historical cloud personalization model $\bar{w}_{r_n}^p$ will be kept at the cloud server r_{cloud} for each RSU server r_n, and therefore a collection of models $\left\{\bar{w}_{r_1}^p, \bar{w}_{r_2}^p, \ldots, \bar{w}_{r_n}^p\right\}$ will be kept on the cloud server r_{cloud}. The attention information of the model $w_{r_n}^p$ uploaded by the RSU server r_n can be expressed as the Euclidean distance between the model and the corresponding historical cloud personalization model $\bar{w}_{r_n}^p$:

$$\xi_{r_n}^p = \frac{\left\| w_{r_n}^p - {}_{r_n}^p \right\|}{\left\| {}_{r_n}^p \right\|} \tag{1}$$

In round p, one personalized global aggregation is performed on the cloud server r_{cloud} for each model $w_{r_n}^p$. The aggregation formula involves the attention information, accuracy and loss of all received models and can be given by

$$\bar{w}_{r_n}^{p+1} = \left(1 - \sum_{i \neq n} \bar{\eta}_{r_i}^p\right) \cdot \bar{w}_{r_n}^p + \sum_{i \neq n} \left(\bar{\eta}_{\eta}^p \cdot \bar{w}_n^p\right) \tag{2}$$

The weights utilized by the model for personalized global aggregation in the RSU server r_i are represented as follows:

$$\bar{\eta}_{r_i}^p = \theta_1 \cdot \frac{\xi_{r_i}^p}{\sum_j \xi_{r_j}^p} + \theta_2 \cdot \frac{acc_{r_i}^p}{\sum_j acc_{r_j}^p} + \theta_3 \cdot \frac{\left(loss_{r_i}^p\right)^{-1}}{\sum_j \left(loss_{r_j}^p\right)^{-1}} \tag{3}$$

where θ_1,θ_2,θ_3 are all fixed values and $\theta_1 + \theta_2 + \theta_3 = 1$.

In round p, the set of models on cloud server r_{cloud} is updated from $\left\{\bar{w}_{r_1}^p, \bar{w}_{r_2}^p, \ldots, \bar{w}_{r_n}^p\right\}$ to $\left\{\bar{w}_{r_1}^{p+1}, \bar{w}_{r_2}^{p+1}, \ldots, \bar{w}_{r_n}^{p+1}\right\}$ after cloud server r_{cloud} performs the above personalization global aggregation for each historical cloud personalization model $\bar{w}_{r_n}^p$. The cloud server r_{cloud} sends each new model to the corresponding RSU server, i.e., the model $\bar{w}_{r_n}^{p+1}$ is sent to the RSU server r_n.

When the RSU server r_n receives the new model $\bar{w}_{r_n}^{p+1}$, it will perform personalized local aggregation of the original RSU model $w_{r_n}^p$. The aggregation formula is as follows:

$$\hat{w}_{r_n}^{p+1} = \left(1 - \eta_{r_n}^p\right) \cdot w_{r_n}^p + \eta_{r_n}^p \cdot \bar{w}_{r_n}^{p+1} \tag{4}$$

where $\eta_{r_n}^p$ is the weight of the personalized local aggregation, which is adaptively obtained by the RSU server r_n based on the model state at this point in time (including attention information $\bar{\zeta}_{r_n}^{p+1}$, model accuracy $a\bar{c}c_{r_n}^{p+1}$ and model loss $lo\bar{s}s_{r_n}^{p+1}$). The attention information can be represented by the Euclidean distance between the cloud personalization model $\bar{w}_{r_n}^{p+1}$ and the original RSU local model $w_{r_n}^p$ can be given by

$$\bar{\xi}_{r_n}^{p+1} = \frac{\left\| \bar{w}_{r_n}^{|p+1} - w_{r_n}^p \right\|}{\left\| w_{r_n}^p \right\|} \tag{5}$$

The attention information of the model $\hat{w}_{r_n}^{p+1}$ obtained after personalized local aggregation is denoted as $\hat{\xi}_{r_n}^{p+1}$, the model accuracy is denoted as $a\hat{c}c_{r_n}^{p+1}$ and the model loss is denoted as $lo\hat{s}s_{r_n}^{p+1}$. The attention information at this point can be represented by the Euclidean distance between the model$\hat{w}_{r_n}^{p+1}$ after personalized local aggregation and the original RSU local model $w_{r_n}^p$:

$$\hat{\xi}_{r_n}^{p+1} = \frac{\left\| \hat{w}_{r_n}^{p+1} - w_{r_n}^p \right\|}{\left\| w_{r_n}^p \right\|} \tag{6}$$

B. Unloading Revenue Model

The task flow focuses on determining the optimal offloading price by considering the utility gained from offloading and the cost incurred, hence the task revenue function is given by

$$R = U - C \tag{7}$$

where U and C represent the utility and cost functions, respectively.

For an application task stream$a \in \mathcal{A}$ waiting to be processed, it must offer an appropriate price p_a^o to incentivize the fleet $\mathcal{V}$ to provide computing resources. Given a pricing strategyp_a^o, each member vehicle$v$ determines the number of data processing unitsn_v^a it is willing to provide to the data stream a.

The task stream $a \in \mathcal{A}$ derives utility from task offloading can be expressed by

$$U_a^o = \sum_{v \in \mathcal{V}} \left[q_v \log_2\left(1 + n_v^a\right)\right]. \tag{8}$$

where q represents the utility coefficient of a unit of computing resources, which quantifies the contribution of each unit of

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computing resources to the utility of the task flow. v represents the amount of computing resources provided by the vehicle, reflecting the actual computing power provided by the vehicle when participating in task offloading.

Based on the amount of computing resources provided by the fleet V, the computational cost of the task stream a is given by

$$C_a^o = \sum_{v \in \mathcal{V}} p_a^o n_v^a \tag{9}$$

where p represents the cost coefficient per unit of computing resources, used to measure the cost of consumption per unit of computing resources. a represents the usage of computing resources, that is, the actual amount of computing resources consumed during task execution. o represents an adjustment coefficient related to the task, taking into account the impact of factors such as task type and priority on cost.

Combining the utility function and the cost function, the benefit function of the task flow a in the unloading phase can be given by

$$\max R_a^o = U_a^o - C_a^o \tag{10}$$

From the perspective of the resource provider v, the utility function that v derives from offloading is given by

$$U_v^o = \sum_{a \in \text{ A}} U_v^a = \sum_{a \in \text{ A}} \left(\delta_v p_a^o N_v n_v^a - n_v^a n_v^a\right) \tag{11}$$

where δ_v and N_v represent the utility level of the vehicle and the maximum allowable supply of computing resources, respectively. Meanwhile, the unit computation cost and V2V data transfer cost are defined as β_v and ω_v. The cost function generated by performing the task is given by

$$C_v^o = \sum_{a \in \mathcal{A}} C_v^a = \sum_{a \in \mathcal{A}} \left(\beta_v \frac{n_v^a}{q_v} + \omega_v n_v^a \sigma\right) d_u \tag{12}$$

Therefore, the revenue received by the fleet member v for participating in the calculation of offloading is given by

$$\begin{aligned} &\max R_v^o = U_v^o - C_v^o \\ &\text{s.t.} \sum_{a \in \mathcal{A}} n_v^a \le N_v, \forall v \in \mathcal{V} \\ &\quad n_v^a \ge 0. \end{aligned} \tag{13}$$

C. Modeling of Recovery Benefits

Given the constant mobility of vehicles, it is essential to recycle and reassign pending tasks from vehicles on the verge of exiting the fleet, ensuring the tasks are completed accurately and thoroughly. To minimize data transfer costs, tasks from departing vehicles will be reassigned, taking into account the available resources of potential providers within the current network.

Calculate the recycling cost of the recycling task $b \in \mathcal{B}$ as p_b^t per unit of calculation, and the recycling cost of b is given by

$$C_b^t = \sum_{v \in \mathcal{V}} \left(\omega_v d_u n_v^b + p_b^t n_v^b\right) \tag{14}$$

where n_v^b represents the number of data processing units provided by v for processing the recycling task b.

The payoff function is given by

$$R_b^t = \sum_{v \in \mathcal{V}} \left[q_v \log_2\left(1 + n_v^b\right)\right] - C_b^t. \tag{15}$$

Analogously to Eqs. (11) and (12), the utility function U_v^t and cost function C_v^t of the vehicle v can be given by

$$\begin{aligned} U_v^t &= \sum_{b \in \mathcal{B}} U_v^b = \sum_{b \in \mathcal{B}} \left(\delta_v p_b^t N_v n_v^b - n_v^b n_v^b\right) \\ C_v^t &= \sum_{b \in \mathcal{B}} C_v^b = \sum_{b \in \mathcal{B}} \left(\beta_v \frac{n_v^b}{q_v} + \omega_v n_v^b \sigma\right) d_u \end{aligned} \tag{16}$$

The unloading treatment benefit of v in the recycling process is:

$$\begin{aligned} &\max R_v^t = \sum_{b \in \mathcal{B}} \left(\delta_v p_b^t N_v n_v^b - n_v^b n_v^b\right) - \sum_{b \in \mathcal{B}} \left(\beta_v \frac{n_v^b}{q_v} + \omega_v n_v^b \sigma\right) d_u \\ &\text{s.t.} \sum_{b \in \mathcal{B}} n_v^b \le N_v, \forall v \in \mathcal{V} \\ &\quad n_v^b \ge 0 \end{aligned} \tag{17}$$

Maintaining a historical cloud personalization model for each RSU server on the cloud overcomes the limited personalization issues found in traditional methods like parameter decoupling and model interpolation, effectively preserving RSU-specific data's local traits. Given the cloud's high computational and storage capabilities, this approach doesn't significantly increase aggregation time, making it both feasible and practical to maintain and update each RSU's model. This strategy improves model information assimilation and personalization for data with local characteristics, leveraging cloud aggregation without major drawbacks.

IV. Joint Incentive-Driven Learning Algorithm Based on Cloud-Edge Game

A. Client-Side and Server-Side Updates

In the first stage, the client uses the global representation part $\mathcal{G}$ received from the server combined with the locally maintained personalization headerH_i to obtain the local model (G, H_i). Let(G, H_i) perform τ_{sync} rounds of training locally to synchronously update the global representation$\mathcal{G}$ and the personalization headerH_i. First calculate and obtain the gradient on the local data$\mathcal{D}_i$ is given by

$$g_i^k = \frac{1}{B} \sum_{j=1}^{B} \nabla \hat{f}_i \left(G_i^k, H_i^k; \xi_{j,i}^k\right) \tag{18}$$

where $\xi_{1,i}^k, \ldots \xi_{B,i}^k \sim D_i$;$B$ is the locally updated batch size. Note that this method uses a smaller batch B to obtain an unbiased estimate of $\tilde{\nabla} f_i\left(G_i, H_i, D_i\right)$ due to the high cost of computing the gradient of the entire batch. A smaller B also improves the generalization ability of the model. In addition, to simplify the algorithmic process,$g_i^k = \nabla''$ is used in the formulation instead of Eq. (18) to represent the process of obtaining the gradient. After obtaining the gradientg_i^k, the global representation G and the personalization header H_i can

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be updated simultaneously. Specifically, for $t = 1, \ldots, \tau_{sync}$, the first stage of the update process can be expressed by

$$\left(G_i^{k,t+1}, H_i^{k,t+1}\right) = \left(G_i^{k,t}, H_i^{k,t}\right) - \eta \cdot g_i^k \tag{19}$$

where η denotes the learning rate of the model locally at the client.

Next, the second stage differs from the first stage in that the global representation G will be frozen and only the personalized header $\mathcal{H}_i$ will be fine-tuned. For $t = \tau_{sync} + 1$ to $t = \tau_{sync} + \tau_{head}$, the fine-tuning process can be expressed by

$$\left(G_i^k, H_i^{k,t+1}\right) = \left(G_i^k, H_i^{k,t}\right) - \eta \cdot g_i^k \tag{20}$$

Set τ_{head} to a value greater than τ_{sync}. Specifically, the FL-CEGID method first synchronously updates the parameters of the global representationG_i and the personalization header H_i, and then fine-tunes the personalization header H_i in multiple rounds. In this way, a smaller τ_{sync} ensures that the global representation part G_i does not deviate too much to the local data distribution after aggregation and cause the model not to converge, while a larger τ_{head} ensures that the local model header has enough personalization capability to improve the local performance of the model.

In the first phase, the server first uses r as a ratio and selects $r \cdot N$ clients to participate in this round of the joint learning process. After that, the server sends the shared global representation G^k after initialization or after the last round of aggregation to the selected clients. The server waits for each client to locally update the global representation G_i^k and the personalized representation H_i^k using the FL-CEGID method.

In the second phase, the server receives the updated global representations G_i^{k+1} returned by each client and aggregates these global representations $G_i^k (i \in N_k)$ to generate a new round of global representation parameters G^{k+1} can be given by

$$G^k = \frac{1}{N_k} \sum_{i=1}^{N_k} G_i^k \tag{21}$$

More importantly, each client will simultaneously upload its updated personalized representation H_i^k. The central server creates and maintains a personalization model head dictionary $\{H_0, \cdots, H_N\}$ and stores the received updated personalization head H_i^{k+1} in the head dictionary. At the end of each training round, the head dictionary will replace the existing personalized head $H_i^{k,\tau}$ with a new personalized head based on the client number index in the manner of Equation (22).

$$\{H_0, \cdots, H_N\} \leftarrow H_i^{k,\tau} \tag{22}$$

where $i \in \{1, \cdots, N_k\}$.

Finally, when the federated learning process ends, FL-CEGID provides each client with a local model consisting of a globally shared representation G and a personalized representation H_i. Each local model can achieve good performance in local data distribution. Meanwhile, FL-CEGID maintains a header dictionary on the server side. By aggregating the personalized headers in the header dictionary, the FL-CEGID method can generate the global header H_{global} with generalization capability, and the generation process can be given by

$$H_{global} \leftarrow \frac{1}{N} \sum_{i=1}^{N} H_i \tag{23}$$

Splice the global header H_{global} with the general global representation G and FL-CEGID ends up providing a global method with generalization capabilities and the ability to perform well in global tests. *B. Incentive Objective Functions and Incentive Drivers* In this paper, utility functions are utilized to quantify the interactions between participants. At the cloud center layer, the edge cloud publishes the total reward r^* to maximize the utility function with the optimization objective of

$$r^* = \arg\max_r U_c\left(d^t, r^t\right) r^* = \arg\max_r U_c\left(d^t, r^t\right) \tag{24}$$

At the edge server layer, based on the total amount of optimal rewards provided by the server r^*, the client decides the training strategy $\left\{d_1^t, \ldots, d_N^t\right\}$, which in turn gives the optimal rewards p for the next stage of data sensing with the optimization objective:

$$\left\langle \left(d_i^t\right)^*, p^* \right\rangle = \arg\max_p U_{e-n}(x, p) + \arg\max_d U_i\left(r^t, d_i^t\right) \tag{25}$$

The user k decides its own data transaction volume and the optimization objective can be given by

$$x^* = \arg\max_x U_k\left(x_k, p\right) \tag{26}$$

The incentives for each participant of the federal learning are achieved through the utility function in the optimization game process described above.

In the middle of the federated learning training process for the cloud-edge game, in order to directly link the incentives to the training process, the process of model distribution is associated with rewards. Specifically, the reward threshold θ is set, $\rho_i = \frac{r_i^t}{\theta} \geq 0$. When $\rho \geq 1$, the server distributes the complete training model for the client i. When $\rho < 1$ is set, the gradient weights are sorted and distributed to the client in a proportional manner. Although such a setting does not proportionally affect the performance of the local model, it is valid for distinguishing the contribution.

B. Solving the Cloud-Edge Game

First determine the existence and uniqueness of the follower game in the cloud-edge game process. For the client i, it is available via $\frac{\partial U_i\left(r_i^t, d_i^t\right)}{\partial d_i^t} = 0$:

$$\frac{\partial U_i}{\partial d_i^t} = \tau r^t \frac{\sum_{j \neq i}^{N} d_i^t - (N-1)\, d_i^{t-1}}{\left(\sum_{i=1}^{N} d_i^t - N d_i^{t-1}\right)^2} - \lambda_i = 0 \tag{27}$$

Solving Eq. (27) yields:

$$\left(d_i^t\right)^* = \sqrt{\tau r^t \frac{\sum_{j \neq i}^{N} d_i^t - (N-1) d_i^{t-1}}{\lambda_i}} + N d_i^{t-1} - \sum_{j \neq i}^{N} d_i^t \tag{28}$$

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$$\sum_{i=1}^{N} d_i^{t^*} = \frac{(N-1)\tau r^t}{\sum_{i=1}^{N} \lambda_i} + N d_i^{t-1} \tag{29}$$

Substituting Eq. (29) into Eq. (28) yields:

$$\left(d_i^t\right)^* = d^{t-1} + \frac{(N-1)\,\tau r^t}{\sum_{i=1}^{N} \lambda_i} - \frac{\lambda_i}{\tau r^t}\left(\frac{(N-1)\,\tau r^t}{\sum_{i=1}^{N} \lambda_i}\right)^2 \tag{30}$$

In the first stage of the cloud-edge game, Eq. (29) can be obtained by substituting Eq. (30) and ordering $\frac{\partial U_c(r^t, d^t)}{\partial r^t} = 0$:

$$\frac{\partial U_c\left(r^t, d^t\right)}{\partial r^t} = \frac{\delta\tau\,(N-1)}{\sum_{i=1}^{N} \lambda_i\left(1 + N d^{t-1}\right) + (N-1)\,\tau r^t} - \kappa \tag{31}$$

The optimal solution for rewarding r^t can be obtained from Eq. (31):

$$\left(r^t\right)^* = \frac{\sigma}{\kappa} - \frac{\sum_{i=1}^{N} \lambda_i\left(1 + N d^{t-1}\right)}{(N-1)\,\tau} \tag{32}$$

The cloud server creates unique personalization models for each RSU and performs global aggregation based on performance metrics like accuracy and loss, enhancing model updates and minimizing variance in federated learning's aggregation. RSUs compute local personalization weights using the same metrics for local updates. The pseudo-code for the federated learning and cloud-edge gaming with an incentive-driven algorithm is as follows:

Algorithm 1 Federated Learning And Cloud-Edge Gaming With Incentive-Driven

Input: Number of clients participating in training per iteration m and local iteration rounds E and reward threshold θ
Output: Global model ω_G

Server:
Initialize global variable$\omega_G$and reward r^0
for $t = 1$ **to**T **do**
  Select $m$available clients from the client set$N$ to form a subset S^t
  Determine the client training strategy for this round$\left(d^t\right)^*$
  for client $i \in S^t$ **do**
    The model size sent to client i based on ρ_i
    $\omega_i^{t+1} \leftarrow localtraining\left(i, \omega_G^t, E, B\right)$
  end for
  $\omega_G^{t+1} = \frac{\sum_{i=1}^{N} |D_i|\omega_i^t}{\sum_{i=1}^{N} |D_i|}$
  Determine the scoper^{t+1} by Calculation
end for

Client:
$localtraining\left(\omega_i, d_i^t\right)$
Data D_i with input size d_i^t
Divide data set $D_i$by B
for $t = 1$ **to**T **do**
  $\omega_i \leftarrow \omega_i - \eta\nabla f_i\left(\omega, B\right)$
end for

V. Experimentation and Analysis

A. Performance Analysis of Onboard Resources for Different Computing Tasks

In this paper, a square area is set up as a simulation area with 15 km on each side, the vehicle can travel normally in an area of such length and the vehicle can communicate effectively. Parameters 100 and 150 indicate that the experimental results are recorded in 100 intervals and 150 intervals, respectively. The communication range between the RSU and the vehicle is 2.2 km. Within this communication range, the vehicle is expected to have enough reaction time to make reasonable judgments and decisions when transmitting information to other vehicles. In order to meet the speed requirement of the vehicle traveling on the highway, it is reasonable to set the maximum and minimum speeds of the vehicle to 50 m/s and 20 m/s, respectively. In order to improve the success rate of data transmission while carrying a certain amount of valid information, the packet size of each unit of data is set to 1.0 k bytes. Due to the short time requirement of vehicle communication, it is desired to transmit data in the shortest possible time slot, the time slot is set to 1.0 ms to fulfill the short time communication. This experimental dataset is obtained from http://www.cvlibs.net/ datasets/kith/.

In order to enable vehicles to perform tasks efficiently and safely, FL-CEGID considers multidimensional incentive mechanism, critical task identification and data interference mechanism. In the multidimensional incentive mechanism, bit rate, matching degree between source and destination nodes, utility function, cost function and social welfare function are considered to motivate more vehicles to participate in the task. After the vehicles participate in the task, they can quickly recognize the critical tasks, which improves the efficiency of completing the task. To secure the mission critical information, data jamming mechanism is used to protect the transmission of mission critical information to the edge server.

This paper uses the following methods to compare performance with FL-CEGID.

Multi Agent Reinforcement Learning (MARL) [29]: This algorithm is a cutting-edge distributed multi-agent algorithm, in which all agents independently train their models locally by only obtaining the same rewards and locally observed environmental information.

Non Orthogonal Multiple Access Mobile Edge Computing (NOMA-MEC) [30]: jointly optimize the calculation unloading and resource allocation of delay and energy consumption by optimizing the objective function and constraints.

Dynamic Threshold Task Offloading Decision (DTTOD) [31] based on dynamic threshold: Vehicle users compare the time consumed by task unloading with a preset dynamic threshold to determine whether to send a task unloading request to the MEC server.

Resource Allocation in Game Theory (RAGT) [32]: In the IoV, the Nash equilibrium of game theory is used to solve the resource allocation problem between Data Service

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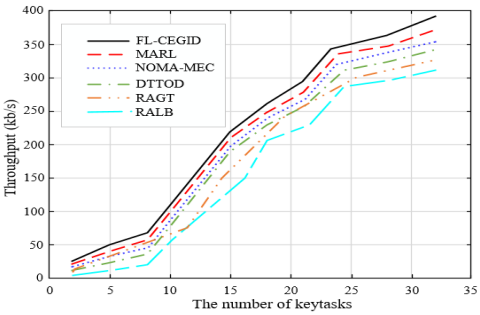


Fig. 2. Throughput at different key tasks.

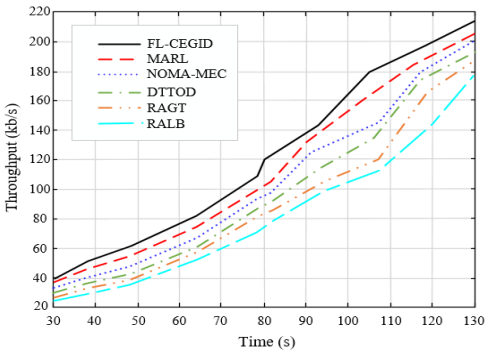


Fig. 3. Throughput at different times.

Operators (DSA) and Fog Nodes (FN), achieving optimal resource allocation.

Resource Allocation for Load Balancing (RALB) [33] based on load balancing: In IoV, by balancing the load of each node, single node overload is avoided, and the overall performance and stability of the system are improved.

In the context of the IoV, due to the high maneuverability of vehicles and their short connection cycles, this paper designs the following experimental indicators to quantitatively analyze algorithm performance, and conducts experimental analysis based on these three indicators.

Throughput (TP): the maximum data rate that the vehicle can receive and forward from the device without dropping frames.

Revenue (RE): The revenue obtained by the source node after the relay node completes the data forwarding.

Satisfaction (SA): Satisfaction of the destination node in obtaining the information after the relay node completes the data forwarding. It is determined by efficient scoring function and then by the score of the node.

The reward value re is determined by the time cost of completing the task and the quality of completing the task. When $tm \geq tm_0$ (tm is the time threshold), the reward value is calculated by $re = -co \cdot tm$, where co is a positive coefficient and tm is the time spent on completing the task, indicating that the reward value increases as the time decreases. When $0 < tm < tm_0$, $re = co \cdot qa$, where qa is the quality of the task being completed, indicating that the higher the quality of the task being completed within a predetermined timeframe, the greater the reward value will be, and the publisher determines the quality of the task. The higher the quality of the task being completed within the specified range, the greater the reward value will be, and the publisher determines the quality of the task. The throughput at different key tasks is shown in Fig. 2, the throughput at different times is shown in Fig. 3.

As shown in Fig. 2, FL-CEGID has the highest throughput, as the number of critical tasks increases, the throughput increases, which leads to an increase in data transfer, the amount of information transferred is positively correlated with the number of critical tasks as well as the throughput. It is clearly evident from the Fig. 3 that the trend of throughput along the time axis and FL-CEGID is still the highest. Since FL-CEGID and the other methods use different auction mechanisms to increase the number of task engagements and completions, the increase in task-related transmissions

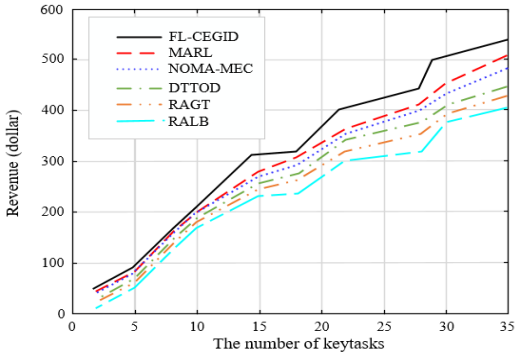


Fig. 4. Revenue at different key tasks.

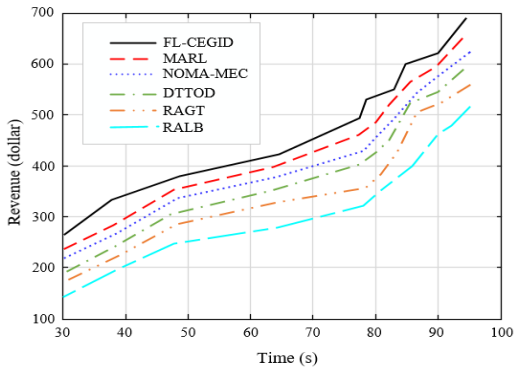


Fig. 5. Revenue at different times.

results in a higher data rate for the vehicle, and therefore, all several algorithms achieve increasing throughput over time. In addition, FL-CEGID designs multidimensional incentives and key task identification methods for vehicular networking, which enable higher data rates for vehicles, ultimately leading to the highest throughput.

Gains increase as the number of critical tasks increases, leading to growth in data communications and consumption of resources. These metrics ultimately lead to an increase in gains. Fig. 4 shows the highest gains. Fig. 5 shows the trend of gains along the time axis.

Through the above experimental results, it can be observed that FL-CEGID gains higher than outperforms other algorithms. FL-CEGID is consistently the highest. Along with the increase in gains, the resources consumed increase over

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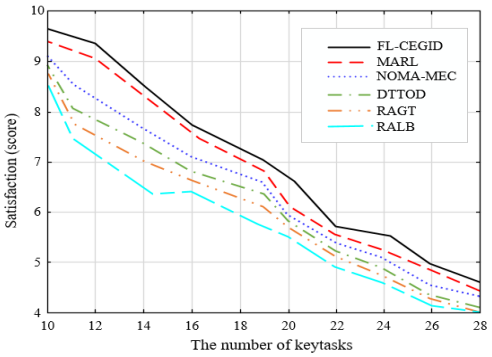


Fig. 6. Satisfaction at different key tasks.

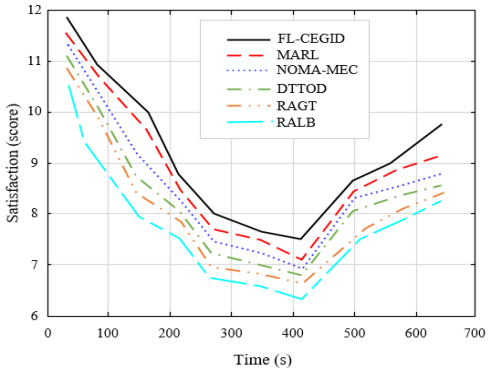


Fig. 7. Satisfaction at different times.

time. The gains will increase as the amount of data transferred between vehicles increases.

As shown in Fig. 6, FL-CEGID performs well on the satisfaction results and as observed the satisfaction decreases when the number of critical tasks increases. As the number of critical tasks increases, the amount of data communication is increasing and more resources need to be consumed. Satisfaction of all algorithms keeps decreasing which is due to the increase in the amount of information conveyed, the number of critical tasks becomes larger and the satisfaction decreases.

FL-CEGID has the highest satisfaction which is due to the fact that multidimensional incentives and critical task identification gradually play a key role in covering the communication range efficiently, and a similar result can be observed in Fig. 7. It is clearly evident from Fig. 7 that FL-CEGID has the highest satisfaction level, the satisfaction level shows a decreasing trend in the first 410 seconds but an increasing trend after 410 seconds. That is because the vehicle is not rewarded for completing the task in the first period and is rewarded due to completing the task in the later period. Rewards are calculated based on three key factors: the quality of the task, the time required to complete such task, and the time spent within the publisher's tolerance. The reward value increases according to the quality of tasks completed within the allowed time frame. Due to the small number of tasks in the initial phase, the vehicle achieved a high level of satisfaction. However, as the amount of transmitted data increases, the satisfaction level changes as time passes and high speed movements bring unknown circumstances and instability.

TABLE I
EXPERIMENTAL PARAMETER SETTINGS FOR COLLABORATIVE MULTI-SOURCE DATA TRANSMISSION

Data source	1	2	3	4
Data size (Mb)	45	35	25	15
Initial position (m)	281.26	1622.54	1137.92	1964.48
Request time (s)	1	5	12	18
Driving direction (E/W)	East	East	West	West

Federated learning architecture inherently protects data privacy by uploading only model parameters, not raw data, to servers for aggregation, significantly reducing data leakage risks. In the IoV context, vehicles train models with their sensor data locally, avoiding raw data exposure. Furthermore, during parameter uploading, homomorphic encryption allows servers to aggregate encrypted parameters without decryption, maintaining data confidentiality. Additionally, digital signature technology verifies parameter source credibility, preventing tampering and ensuring data integrity and authenticity.

B. Multi-hop Collaborative Transmission Experiment

In order to realize the "store-carry-forward" mechanism in IoV, this paper conducts multi-hop collaborative transmission experiments, and the information transmission experiments with a single data source point are used to check the accuracy of the transmission capacity and transmission delay analysis. In the multi-source data transmission experiments, different locations and data file sizes of transmission sources are considered, the transmission delay and system communication resource consumption between the proposed strategy and other control methods are compared. The parameters are set as follows, vehicle density: 2.5-25 vehicles/km, roadside cell spacing: 2 km, vehicle speed: 25 m/s, transmission mode switching period: 55 ms, relay node selection period: 2 sec. In the IEEE 802.11 protocol, multiple transmission data rates can be achieved by selecting different modulation types depending on the communication state. Therefore, in each cycle of the control experiment, transmission data rate randomly takes one of 0.5 Mbps, 1 Mbps and 1.5 Mbps with equal probability. The experimental results show that when the real-time transmission data rate is varied, the multi-hop transmission performance of the information is slightly degraded compared to the fixed data rate. This is because the time-varying transmission data rate leads to an imbalance of data sending in different transmission links, i.e., there is not enough data available for transmission in some quality links, while some of the data will be congested in the links with poorer status. The specific setup of the multi-source data transmission experiment is shown in Table I.

In the multi-source data transmission experiment, four different data sources initiate data transmission requests at different locations and at different moments. The data source vehicles and non-data source vehicles utilize FL-CEGID real-time collaborative transmission as well as several other transmission methods for data transmission, respectively. The experiment counts the delay required for each data source

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TABLE II
MULTI-SOURCE DATA TRANSMISSION DELAY UNDER DIFFERENT TRANSMISSION STRATEGIES (UNIT: SEC)

Transmission strategy	Data source 1	Data source 2	Data source 3	Data source 4
FL-CEGID	62.1	28.8	40.5	50.6
MARL	69.4	30.2	47.8	77.3
NOMA-MEC	72.6	34.5	49.6	80.8
DTTOD	75.2	36.4	52.7	83.2
RAGT	81.6	38.3	56.7	84.9
RALB	84.3	39.7	61.8	87.4

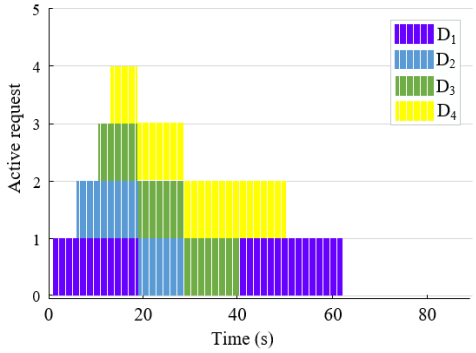


Fig. 8. Resource utilization of FL-CEGID.

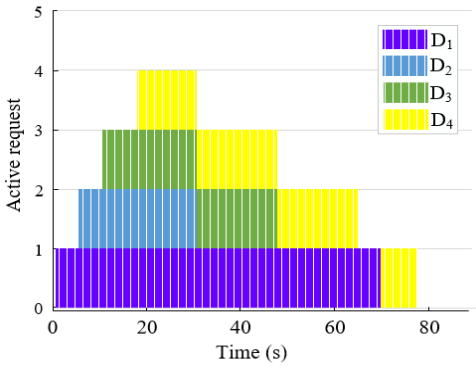


Fig. 9. Resource utilization of MARL.

to complete the data transmission and the system resource utilization during the transmission process. The experimental results are shown in Table II.

It is clearly evident from Table II that FL-CEGID can achieve shorter transmission delay than several other methods. This is because FL-CEGID transmission strategy always selects sources that can achieve higher transmission data rates, whereas the other methods need to search for potential relay nodes and transmission paths in the absence of control. Fig. 8 shows resource utilization of FL-CEGID.

FL-CEGID also maintains better fairness because each source point can always reach the optimal location for data transmission at a certain time period. The access control mechanism in algorithms restricts access to data and models. Fig. 9 shows resource utilization of MARL.

Only authorized and legitimate vehicles and servers can participate in the federated learning process and obtain relevant data and model information. Through identity authentication

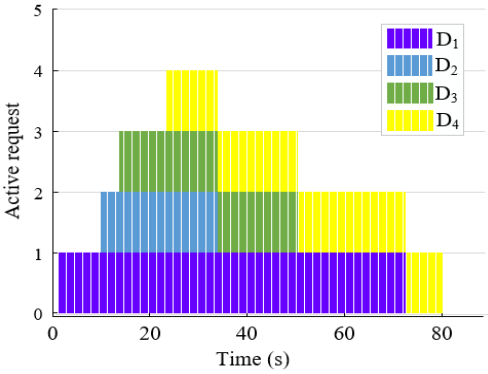


Fig. 10. Resource utilization of NOMA-MEC.

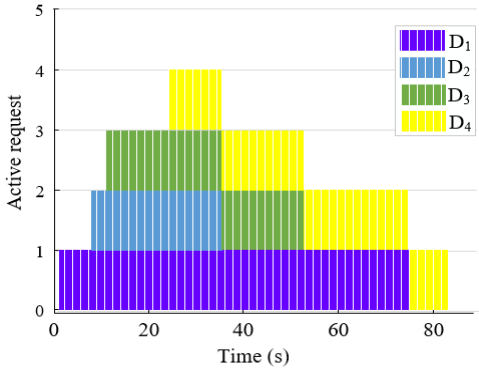


Fig. 11. Resource utilization of DTTOD.

technology, vehicles and servers are authenticated to ensure that only trusted nodes can access the system. Fig. 10 shows resource utilization of NOMA-MEC.

In the process of data storage and processing, fine-grained access permissions are set according to different user roles and task requirements. Vehicles can only access data related to their own tasks, while edge servers and cloud servers access corresponding levels of models, data according to their respective responsibilities to prevent unauthorized access and data abuse, comprehensively protect data privacy and security. Fig. 11 shows resource utilization of DTTOD.

It can be observed from Fig. 11 that FL-CEGID can greatly reduce the resource occupation and transmission delay. This is because the method guarantees a non-competitive transmission scheduling mechanism, i.e., any relay node or RSU will not be used for data transmission by more than one source point at the same time, thus enhancing the information transmission efficiency. Fig. 12 shows resource utilization of RAGT.

Since the transmission capacity analysis and the delay analysis take into account the data volume of the message, the transmission location, and the vehicle traffic data, only the source point at the appropriate location will be allowed to transmit the data instead of relying on the proximity alone to select the transmission source point. Fig. 13 shows resource utilization of RALB.

Although the FL-CEGID algorithm performs well in throughput, revenue, and satisfaction metrics, there is still room for improvement in resource allocation efficiency in

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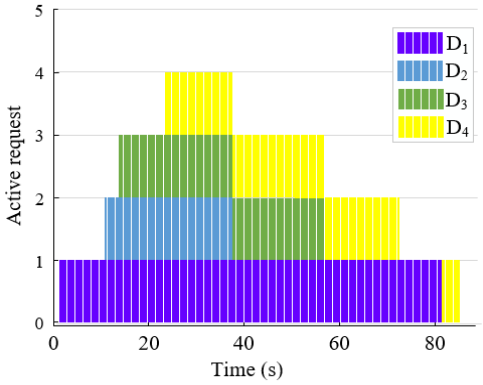


Fig. 12. Resource utilization of RAGT.

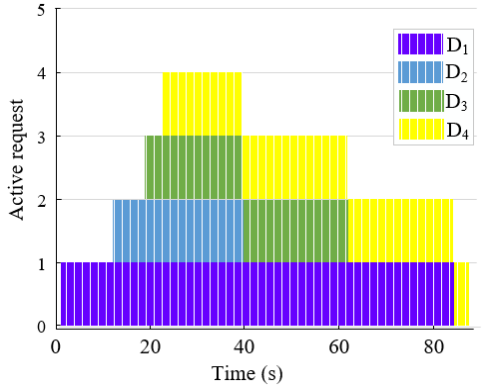


Fig. 13. Resource utilization of RALB.

high load network environments, such as scenarios with high vehicle density and frequent task requests. Further optimization of incentive mechanisms can be considered to enable vehicles to allocate computing resources more reasonably and avoid performance degradation caused by resource competition. In addition, to address the latency issue of algorithms in handling large-scale data transmission, research is being conducted to optimize data transmission strategies, such as adopting more efficient encoding methods or caching mechanisms, in order to improve data transmission efficiency and reduce latency.

VI. Conclusion and Future Work

Traditional optimization algorithms are far from being able to meet the needs of resource allocation and computational task offloading in IoV due to model curing and large computational complexity. This paper adopts the federated learning approach to address the challenges of resource uncertainty, resource dynamics, and high concurrency of multitasking in vehicular network, such as the lack of environment awareness, environment instability, and high complexity. The experimental results verify the accuracy of the theoretical analysis results, and also show that FL-CEGID can effectively improves the data transmission performance and reduces the occupancy of the system communication resources, improves the satisfaction and reliability level, respectively, which not only has good convergence performance and practicability, but also has better offloading benefits in different scenarios.

The future work direction focuses on optimizing algorithm performance and improving model aggregation strategies. This involves exploring more advanced model aggregation algorithms, such as those based on dynamic weight adjustment or model compression techniques, to further enhance the precision and training efficiency of federated learning models, and to reduce the impact of vehicle data distribution differences and high mobility on model convergence.

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